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Thm (Euclid) There are infinite many primes.

Idea: $p_1, \cdots, p_k$ are the only primes

$\Rightarrow n = p_1 \cdots p_k + 1 > 1$ is indivisible by $p_1, \cdots, p_k$.

$\Rightarrow$ contradiction (n has a prime factor $q \neq p_1, \cdots, p_k$)

Variations:

Thm 1. There are infinite many primes $\equiv 3 \pmod 4$.

Adapting Euclid's proof: $p_1, \cdots, p_k$ are the only primes $\equiv 3 \pmod 4$

$\Rightarrow n = 4p_1 \cdots p_k - 1 > 1$ is indivisible by $p_1, \cdots, p_k$.
$\equiv 3 \pmod 4$

$\Rightarrow$ contradiction (n has a prime factor $q \neq p_1, \cdots, p_k$,
$q \equiv 3 \pmod 4$)

Thm 2. There are infinite many primes $\equiv 5 \pmod 8$ (which are also $\equiv 1 \pmod 4$).

Idea: $p_1, \dots, p_k$ are the only primes $\equiv 5 \pmod 8$

$\Rightarrow n = \underline{\quad ? \quad} > 1$ is indivisible by $p_1, \dots, p_k$.

$\equiv 5 \pmod 8$

$\Rightarrow$ contradiction (n has a prime factor $q \neq p_1, \dots, p_k$, $q \equiv 5 \pmod 8$)

Want $q \mid n \Rightarrow q \equiv 1 \pmod 8$

or $q \equiv 5 \pmod 8$

Issue: A prime $q \mid n$ can still be $\equiv 1, 3, 5, 7 \pmod 8$.

$\equiv 3 \pmod 4$

E.g. $3 \cdot 7 \equiv 5 \pmod 8$.

Recall: If $p \equiv 3 \pmod 4$ and

$$p \mid n = x^2 + y^2,$$

then $p \mid x$, $p \mid y$.

Takeaway: $p \nmid x$ or $p \nmid y$ $(p \equiv 3 \pmod 4)$

$$\Rightarrow p \nmid n = x^2 + y^2.$$

In particular, if $(x, y) = 1$, then

$$n = x^2 + y^2$$

is indivisible by any prime $\equiv 3 \pmod 4$.

Try $n = \underbrace{(p_1 \cdots p_k)^2}_{\equiv 1 \pmod 8} + 2^2 > 1$ is indivisible by $p_1, \ldots, p_k$

$\equiv 5 \pmod 8$

But n has a prime factor q!

① $q \neq p_1, \cdots, p_k$

② $(p_1 \cdots p_k, 2) = 1 \Rightarrow q \not\equiv 3 \pmod 4$

$\Rightarrow q \not\equiv 3, 7 \pmod 8$.

$\Rightarrow q \equiv 1, 5 \pmod 8$.

③ A product of primes $\equiv 1 \pmod 8$ is $\equiv 1 \pmod 8$.

So n must have a prime factor $q \equiv 5 \pmod 8$.

Thm (Dirichlet) Let $q \geq 1$ and $a \in \mathbb{Z}$ be integers s.t. $\gcd(a, q) = 1$. Then there are infinitely many primes $p \equiv a \pmod q$.

Why do we need $\gcd(a, q) = 1$?

Look at $q \cdot m + a$. If $p \mid q$, $p \mid a$, then

$$p \mid qm + a.$$

But in order for $qm + a$ to be prime, we must have

$$p = qm + a.$$

There is at most 1 choice of m s.t. $qm + a$ is prime.

Euler's analytic proof.

UFT in $\mathbb{Z}$: $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$ uniquely up to order.

Look at

$$\infty = \sum_{n=1}^{\infty} \frac{1}{n} \overset{?}{=} \prod_p \left(1 + \frac{1}{p} + \frac{1}{p^2} + \frac{1}{p^3} + \cdots\right)$$

$\frac{1}{1-\frac{1}{p}} = (1-p^{-1})^{-1}$

p: prime

E.g. $n = 300 = 2^2 \cdot 3 \cdot 5^2$.

$$\frac{1}{n} = \frac{1}{2^2} \cdot \frac{1}{3} \cdot \frac{1}{5^2} : \left(1 + \frac{1}{2} + \frac{1}{2^2} + \cdots\right)\left(1 + \frac{1}{3} + \cdots\right)\left(1 + \frac{1}{5} + \frac{1}{5^2} + \cdots\right)$$

$\frac{1}{2^2} \cdot \frac{1}{3} \cdot \frac{1}{5^2}$

Conclusion:

$$\sum_{n=1}^{\infty} \frac{1}{n} = \prod_p \left(1 + \frac{1}{p} + \frac{1}{p^2} + \frac{1}{p^3} + \cdots\right).$$

Will prove next time, by truncating $\prod_p$, that

Thm (Euler) The series

$$\sum_{p} \frac{1}{p}$$

diverges. In particular, there are infinitely many primes.